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Technology of Metallic Materials A.Y. 2025-26

Mechanical Engineering Course



Corrosion

G. Maizza
DISAT

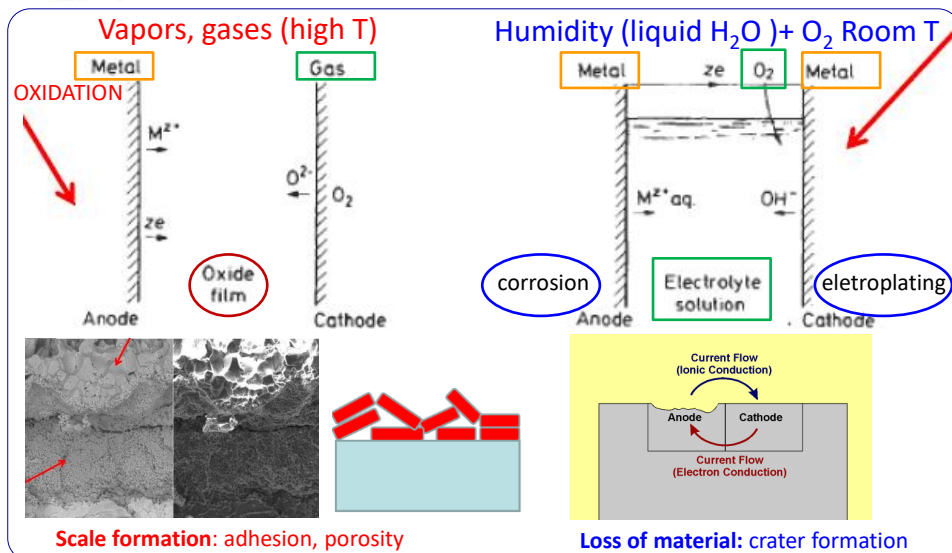
Lecture 11

1



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MAIN TYPES OF CORROSION DRY and WET CORROSION



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2

2



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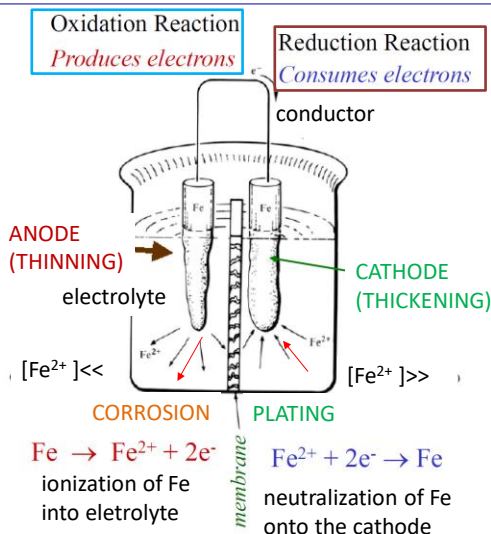
CORROSION IN IRON AND STEELS

BASIC PHENOMENA OF WET CORROSION

- The **anode** gives up electrons to the circuit and corrodes.
- The **cathode** receives electrons from the circuit by means of a chemical or cathode reaction. By-products may form.

Anode, cathode, conductor and electrolyte must be electrically connected to promote corrosion

In general the moisture near the metallic parts, in the presence of the oxygen in the environment, causes corrosion.



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3

3



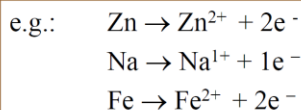
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ANODIC REACTION

OXIDATION REACTION (occurs at the ANODE) - Corrosion

Metals $\rightarrow$ Ions

$\text{M} \rightarrow \text{M}^{n+} + n\text{e}^-$



Metallic portions of the anode electrode precipitate as ions into the electrolyte solution (whereas the released electrons flow through the conductor towards the cathode)

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CATHODIC REACTION

The type of reactions depends on the local condition at the cathode:

- ✓ If H^+ ions are available then: $2H^+ + 2e^- \rightarrow H_2$
- ✓ If acid solution with dissolved oxygen: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$
- ✓ If basic/neutral with dissolved oxygen: $O_2 + 2H_2O + 4e^- \rightarrow 4(OH^-)$

• If some metal ions are present they may be reduced by the electrons:



Can get more than one of these occurring at the same time

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5



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Classification of Corrosion Phenomena

UNIFORM C.



GALVANIC C.



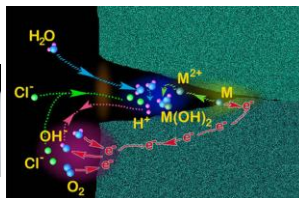
PITTING C.



CREVICE C.



CREVICE C.



6



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Other CORROSION PHENOMENA OCCURRING SYNERGISTICALLY

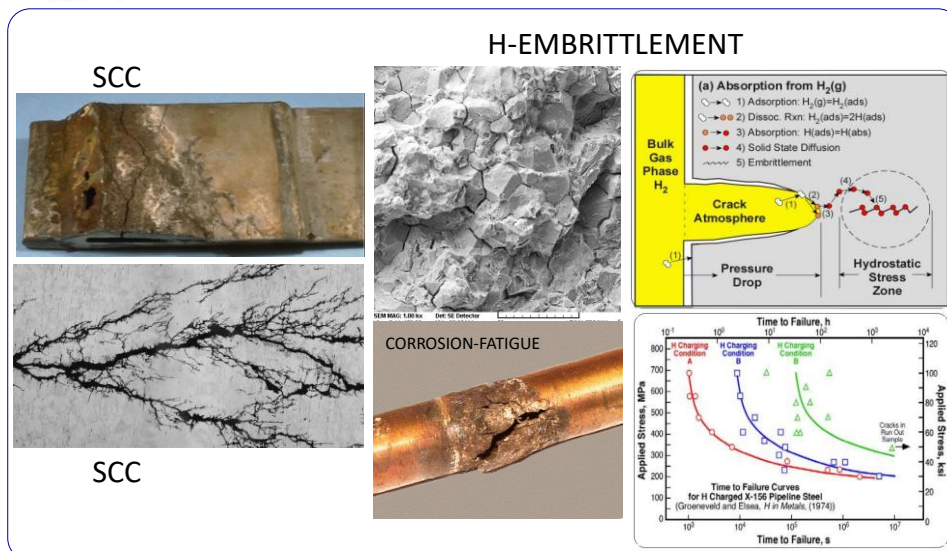
- **Stress Corrosion Cracking** – enhanced by external loads
- **Hydrogen Embrittlement** – enhanced by the presence of H (from external sources or formed in-situ during corrosion, welding, etc.)
- **Corrosion Fatigue** - attack is affected by mechanical factors
- **Selective Leaching** - one element of the alloy is removed by corrosion.
- **Fretting Corrosion** - low frequency sliding + corrosion
- **Erosion Corrosion** - attack is modified by flow effects.
- * **Microbiological Corrosion** (ex. biomedical prosthesis, water pipes. Etc.)

7



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Synergistic Phenomena



8



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UNIFORM CORROSION

- **Relevance:** responsible for the **greatest destruction of structural materials**, but technically **not a major cause for concern**
- **Controlling factor:** the **relative extent** of the anodic and cathodic areas
- **Mechanism:** corrosion cells are very small, numerous, and randomly distributed over the surface of the metal, such that the effect is **more or less of uniform attack on the surface**
- **Rate of corrosion:** proportional to the flow of electron current which may depend on a number of factors but **the important criterion** is the balance between the rates of anodic and cathodic reactions required to maintain electro-neutrality (whichever reaction occurs least readily will control the corrosion rate).

9



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GALVANIC or BIMETALLIC CORROSION

- Two dissimilar metals in relative contact and immersed in a conducting solution usually develop **different corrosion potentials**
- Such potential difference is the **driving force** for corrosion:
 - the less noble of the two metals corrodes more rapidly than the other
- A standard **electrode potential table**, based on standard galvanic series data for the metals with respect to H potential, is commonly used for electrochemical ranking of the metals:
 - the **table** shows which alloy of the selected galvanic couple tends to be either anodic or cathodic
 - however, **the tables give no indication on the rate of corrosion**
- **Surface passivity** of metals markedly affects the corrosion rate, by decreasing it, but this effect is not taken into account in the table

10



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STANDARD ELECTRODE POTENTIALS (CONDITIONS: 300 K, MOLAR SOLUTION)

Oxidation reaction for solution of the metal	Normal hydrogen scale (volts)
$\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^-$	- 2.36
$\text{Al} \rightarrow \text{Al}^{3+} + 3\text{e}^-$	- 1.66
$\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$	- 0.76
$\text{Cr} \rightarrow \text{Cr}^{3+} + 3\text{e}^-$	- 0.74
$\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$	- 0.44
$\text{Ni} \rightarrow \text{Ni}^{2+} + 2\text{e}^-$	- 0.25
$\text{Sn} \rightarrow \text{Sn}^{2+} + 2\text{e}^-$	- 0.14
$\text{Pb} \rightarrow \text{Pb}^{2+} + 2\text{e}^-$	- 0.13
$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$	0.00
$\text{Sn}^{2+} \rightarrow \text{Sn}^{4+} + 2\text{e}^-$	+ 0.15
$\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$	+ 0.34
$\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4(\text{OH})^-$	+ 0.40
$\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$	+ 0.77
$\text{Ag} \rightarrow \text{Ag}^+ + \text{e}^-$	+ 0.80
$2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$	+ 1.23
$\text{Au} \rightarrow \text{Au}^{3+} + 3\text{e}^-$	+ 1.42

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11

11



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FACTORS AFFECTING Galvanic Corrosion

- **Environment/Separation effect**
 - galvanic effects are reduced in proportion to the **distance from the cathodic region** and to the **resistivity** of the environment
 - the attack at the anodic regions is sensibly reduced in **poorly conducting solutions**, where the electric resistance of the corrosion cell plays a major role in controlling the corrosion current
- **Ratio of the Surface areas effect.**
 - is the key factor used to prevent galvanic corrosion by design
 - if a large cathodic area is near to a small anodic area (with surface ratios of hundreds or even thousands), then the rate of attack of the anodic material may become extremely high indeed.
 - Conversely, large surface anodes coupled to small surface cathodes may not undergo a sluggish corrosion.
- **Temperature of the environment**
- **PH of the environment**

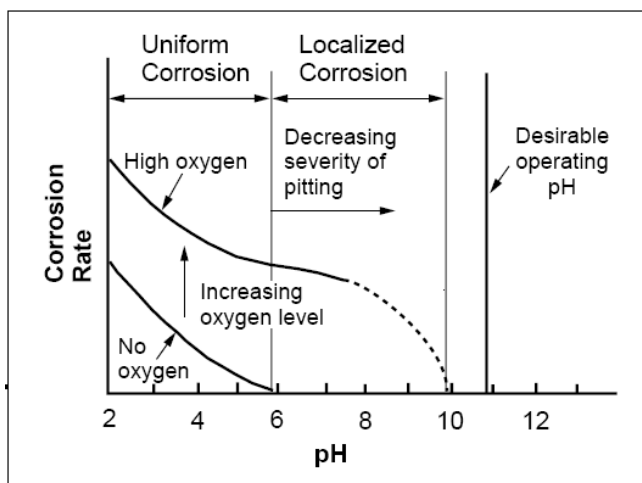
12



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FACTORS CONTROLLING CORROSION RATE OTHER THAN ELECTRODE POTENTIAL

- % O in solution
- PH
- Temperature



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13

13



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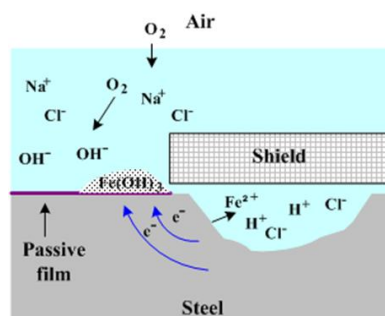
Localized Corrosion or Crevice Corrosion

-Favored by the presence of crevices, either as a result of (wrong) **design** or by formation of **deposits**, or the like

- Such obstructed **regions** in which oxygen, the usual cathodic reactant, cannot be replenished.

-Whereas both anodic and cathodic reactions once occurred uniformly over the surface, after a short period **the only reaction remaining in the crevice may be the anodic reaction**, balanced by **cathodic action outside the crevice**.

-It is **most serious in active/passive materials** where a sequence of events may follow leading to significant and irreversible changes in the environment within the crevice.



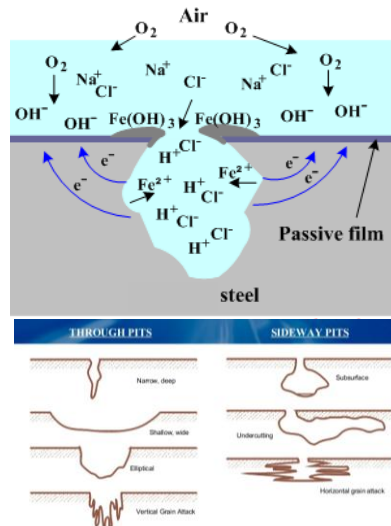
14



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Localized Corrosion: Pitting Corrosion

- **Extremely dangerous** because it is localized whereas **the overall metal loss is negligible**.
- It occurs predominantly as a result of the **action of chloride ions**, although **bromides, hypochlorites and a few other species are occasionally responsible**, and it is particularly insidious because it frequently occurs in **environments which are close to provide complete protection** for the alloy except for occasional fluctuations in conditions.
- The phenomenon can cause **major changes of the environment within the pit**, when an **autocatalytic increase in corrosion rate** can occur



15



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H embrittlement

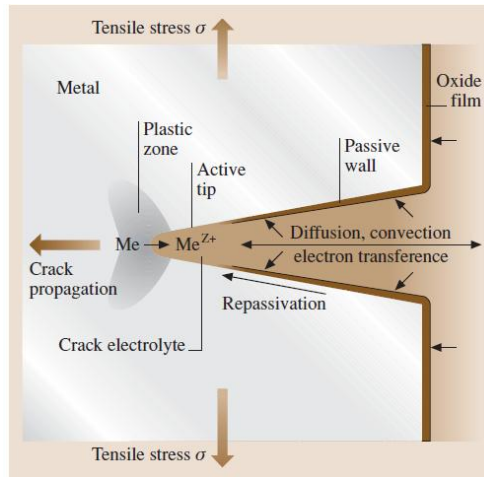
- Hydrogen embrittlement is a mode of **low ductility fracture** which may be induced **in ferritic and martensitic steels at high strength levels** and in **titanium alloys** by the introduction of hydrogen.
- Although it is relatively simple to demonstrate the distinction between hydrogen embrittlement and true stress corrosion in suitable laboratory tests it is by no means easy to separate the mechanisms in practical failure situations with high-strength materials.
- Thus for example much of the stress-corrosion cracking reported in alloy steels in **hydrogen sulphide** or **cyanide** environments is undoubtedly due to the effects of hydrogen rather than to anodic processes.
- The hydrogen may arise from a **variety of sources** including electroplating and welding, but the problem is of greater concern when the hydrogen is generated by corrosion.
- Not only is this continuously available while the alloy is exposed, but for sufficiently susceptible alloys the only environmental requirement is the presence of water in some form.

16



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STRESS CORROSION CRACKING



Scheme of stress corrosion crack propagation in a passive metal by quasi continuous, intermittent crack tip- slip activation followed by deformation-enhanced dissolution of metal at the crack tip. Note the role of a plastic deformation zone at the crack tip and protection of the crack walls by repassivation, after crack growth has occurred

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17

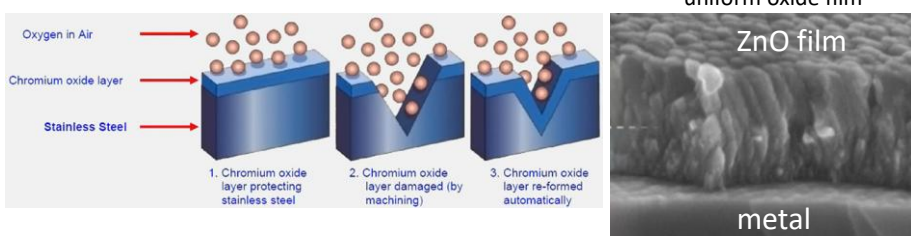
17



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The Passivating Oxide film

- Many metals display active corrosion behaviour when immersed in a corrosive solution
- Others metals including many engineering alloys carry a stable film of solid corrosion product, frequently an **oxide**.
- If this film has **good cohesive strength and adhesion to the metal**, and **low ion conductivity**, the **rate of metal dissolution is limited** to the rate at which metal ions can pass through the film, often a factor of $10^3 - 10^6$ times less than the rate of corrosion of uncoated metals.
- This is the phenomenon of **passivity**, exhibited by **stainless steels**, **nickel alloys**, **titanium** and many similar materials and responsible for their utility as engineering materials in corrosive environments



18



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Prevention of Corrosion

1. Weld rather than rivet
2. Containers should be designed for easy draining
3. Design for easy replacement of expected failures
4. Avoid stresses in corrosion exposed components
5. Avoid electrical contact between dissimilar materials
6. Avoid heterogeneity (metals, vapor spaces, heat, stress)
7. Multiphase and particularly lamellar structure materials are prone to corrosion (exception is duplex S.S.)

19



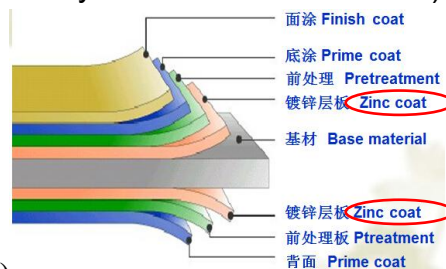
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Protection Methods

In the case of steels, coating the anode system with **Zn** or **Mg** will save the mechanical component from corrosion (ex. in galvanized sheet steels Zn or Mg sacrifice themselves to save Fe)

Coatings

- metallic – often dual function
- cladding
- flame spraying, vapor deposition
- electro-deposition (electroplating)
- hot-dipping (Zn hot dip coating)
- polymers

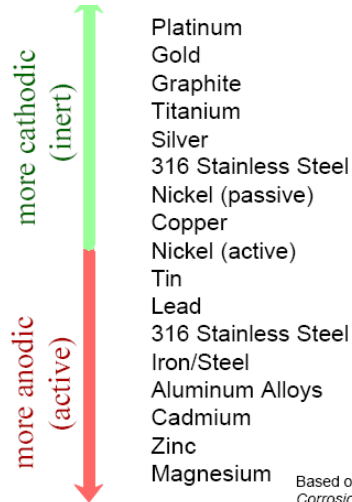


20



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Ranks Of reactivity of Metals/Alloys in seawater



More realistic and practical ranking.

Alloys are the **top are cathodic** and unreactive, whereas those at the **bottom are most anodic**.

No voltages are provided.

Based on Table 17.2, *Callister 6e*. (Source of Table 17.2 is M.G. Fontana, *Corrosion Engineering*, 3rd ed., McGraw-Hill Book Company, 1986.)